

Exact Fourier certificates for complex Hadamard matrices of order six

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Abstract

We prove the Fourier-vanishing conjecture of Matolcsi, Ruzsa, and Weiner: every complex Hadamard matrix of order six outside Tao’s cubic equivalence class has a vanishing character sum at every permutation of $(1, 1, 1, -1, -1, -1)$. We also give an exact certificate excluding seven mutually unbiased bases in $\mathbb{C}^6$; by Weiner’s completion theorem, this yields an upper bound of five. Both certificates use only integer and rational arithmetic, and the complete verifier is included.

1 Introduction

A complex Hadamard matrix of order six is a matrix $H \in \mathbb{T}^{6 \times 6}$ with $H^*H = 6I$, where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. Its columns become an orthonormal basis after normalization by $1/\sqrt{6}$. The same columns are also points of the torus $\mathbb{T}^6$, and their character sums record constraints that orthogonality places on their phases. We study the normalized sums

$$g_H(a) = \frac{1}{6} \sum_{k=0}^5 \prod_{i=0}^5 H(i, k)^{a_i}, \quad a \in \mathbb{Z}^6, \quad \sum_{i=0}^5 a_i = 0. \quad (1)$$

The balancing condition makes these sums invariant under multiplying individual columns by phases. Write $e_0, \dots, e_5$ for the standard coordinate vectors. Orthogonality gives $g_H(e_i - e_j) = 0$ for $i \neq j$. A more restrictive vanishing at the charge

$$\alpha = (1, 1, 1, -1, -1, -1)$$

was conjectured by Matolcsi, Ruzsa, and Weiner [15, Conjecture 2.3]. It links a degree-six character to the structure of an individual Hadamard matrix, without assuming that the matrix belongs to a larger family of unbiased bases.

The conjecture has one exceptional equivalence class. Here two Hadamard matrices are equivalent if they differ by row and column permutations and by multiplication of individual rows and columns by elements of $\mathbb{T}$. Put $\omega = \exp(2\pi i/3)$, and define $T = (\omega^{t_{ij}})_{i,j=0}^5$ by

$$t_{ij} = \begin{cases} 0, & i = 0 \text{ or } j = 0 \text{ or } i = j, \\ 1, & i, j \in \{1, \dots, 5\}, \quad i - j \equiv \pm 1 \pmod{5}, \\ -1, & \text{otherwise.} \end{cases} \quad (2)$$

This is the cubic matrix in Tao’s construction [23, p. 254]; the same representative already appears in Moorhouse’s study of 2-transitive complex Hadamard matrices [18, Example 1.1]. Its five noninitial indices form a cycle: adjacent indices carry the exponent 1 and nonadjacent distinct indices carry -1 . The matrix and its entrywise square are both Hadamard.

Theorem 1.1 (Fourier vanishing). *Let H be a complex Hadamard matrix of order six. If H is not equivalent to the cubic matrix T in (2), then*

$$g_H(\pi\alpha) = 0 \quad \text{for every coordinate permutation } \pi.$$

Theorem 1.1 proves the conjecture for arbitrary complex Hadamard matrices. The normalization in (1) divides the character sums used in [15] by six and therefore leaves the vanishing assertion unchanged. Multiplying rows by phases multiplies each character by a unit scalar; together with the balancing condition, this shows that the assertion respects the stated equivalence relation.

1.1 Fourier methods and mutually unbiased bases

Two orthonormal bases in $\mathbb{C}^d$ are mutually unbiased if every inner product between their vectors has modulus $d^{-1/2}$. Such bases describe complementary measurements and arise in quantum state determination [7, 25]. A family contains at most $d + 1$ bases, and complete families attaining that bound exist in prime-power dimensions [25]. The first dimension outside that construction is six. Write $N(6)$ for the maximum size of a mutually unbiased family in $\mathbb{C}^6$.

Fixing one basis identifies every other unbiased basis with a complex Hadamard matrix. Matolcsi developed a Fourier and Delsarte approach to the resulting constraints on torus points [13]. Matolcsi, Ruzsa, and Weiner developed identities for the associated character sums and formulated the vanishing conjecture addressed here [15]. Maxwell and Brierley proved that vanishing for the Karlsson family [17, Theorem 4.1]. Theorem 1.1 removes the family restriction. The exceptional matrix is usually denoted S_6 in the Hadamard literature [22, Section 5.6.4]; Section 4 gives the precise identification with (2).

The structural step has a separate antecedent. Lisoněk introduced *S-Hadamard matrices*, for which both H and $H^{\circ 2}$ are Hadamard [12, Definition 2.1]. Phangara conjectured that every order-six S-Hadamard matrix is equivalent to the cubic example [21, Conjecture 3.1.6]. Proposition 4.1 proves this assertion. The uniqueness of the cube-root class itself appears in the Butson classification of Lampio, Östergård, and Szöllősi [10, Theorem 4.1 and Table 2] and in Liang, Chen, Long, and Qiu [11, Lemma 10]. The additional step here derives cube-root entries from the two Hadamard conditions on arbitrary phases.

There is also a substantial history of positive certificates for the MUB problem. Brierley and Weigert formulated polynomial and semidefinite approaches [3]. Matolcsi and Weiner strengthened the Delsarte witness method for prescribed Hadamard families [16], and Kolountzakis, Matolcsi, and Weiner developed the positive-definite-function approach on compact groups [9]. Gribling and Polak applied permutation symmetry to positive moment matrices in a tracial noncommutative formulation of the MUB problem [6]. Our finite invariant-moment certificates belong to this line of work. Their specific character identities and shared-column test functions are described below.

Bandeira, Doppelbauer, and Kunisky proved that positive-definite polynomials of degree at most six in the entries of a single unitary transition matrix and their conjugates cannot improve the seven-basis bound in dimension six [1, Theorem 1.10]. That result concerns its specified single-matrix relaxation. The mixed certificate here uses products of character sums from several transitions and relations of higher total order.

The individual-matrix statement also supplies an input to a second exact certificate within this paper. It yields the following complete-family obstruction.

Theorem 1.2 (No complete family). *There do not exist seven orthonormal bases $B_r = (b_{r,0}, \dots, b_{r,5})$ of $\mathbb{C}^6$, $1 \leq r \leq 7$, such that*

$$|\langle b_{r,i}, b_{s,j} \rangle|^2 = \frac{1}{6} \quad (r \neq s, 0 \leq i, j \leq 5).$$

By Weiner’s completion theorem, any d mutually unbiased bases in $\mathbb{C}^d$ extend to a complete family [24, Corollary 3.4]. Theorem 1.2 therefore gives $N(6) \leq 5$. Under its stated binary64 arithmetic and compiler

conditions and complete-execution certificate contract, the companion paper [19, Theorem 1.1, Section 7 and Appendix A] gives the sharper value $N(6) = 3$. The Fourier theorem concerns each individual Hadamard matrix; its content is distinct from a bound on the number of bases. For transitions in MUB triplets, Matolcsi, Matszangosz, Varga, and Weiner conjecture both this vanishing and a further condition coupling characters at 3α for H and H^* [14, Conjecture 2]. Their route to $N(6) = 3$ also uses a separate conjecture excluding the Szöllősi family from quadruplets [14, Conjecture 3 and Proposition 4.2].

1.2 Proof strategy

The main difficulty is to obtain a universal statement about phases from orthogonality without restricting the matrix to a parametrized family. We average monomials in its entries over distinct row and column indices. Summing a row with one exponent $+1$ and one exponent -1 against orthogonality expresses its moment in terms of moments with merged rows. A formal seven-row version has no unused physical row; it gives a homogeneous relation among six-row moments. Products of character sums expand into these injective moments by grouping column samples according to which indices coincide.

The first certificate adds an auxiliary absolute square to a product $|g_H(\alpha)g_H(q)|^2$ and proves that their finite average is zero. Every summand is nonnegative, so the product vanishes for every row ordering. Two choices of q force a dichotomy: either all permuted α characters vanish, or the entrywise square $H^{\circ 2}$ is Hadamard. In the latter case, Newton’s identities constrain every row ratio to two rotated triples of cube roots. An orthogonality argument then forces the whole dephased matrix to be cubic, and a five-cycle classifies it as T . This proves Theorem 1.1.

The complete-family obstruction requires more than the vanishing condition on one matrix. We first exclude a cubic transition matrix from such a family by an exact finite character calculation. For the remaining case, we average products of character sums over ordered basis pairs and triples. Sampling one additional column shared by all test functions expresses their Gram entries as combinations of three-factor moments. Six explicitly specified positive quadratic forms have a combined expansion whose constant term dominates, with negative sign, the sum of the absolute values of all other coefficients. Since every retained moment has absolute value at most one, this contradicts positivity.

The certificates are finite rational identities. Modular elimination locates candidate substitutions, and exact integer multiplication proves that they hold for every actual moment vector. The calculation does not require those moment values to be rational. Section 7 explains this distinction and traces the verifier’s operations back to the sampled variables.

Section 2 fixes the character notation. Section 3 constructs the single-matrix certificate and its dichotomy. Section 4 classifies the cubic alternative and proves Theorem 1.1. Section 5 derives the complete-family identities and excludes a cubic pair. Section 6 constructs the shared-column Gram certificate and proves Theorem 1.2. Section 7 verifies both exact calculations, and Appendix A prints the complete standard-library Python verifier, including all integer weights.

2 Characters and phase invariance

The Hermitian inner product is conjugate-linear in its first argument. We index the coordinates of $\mathbb{C}^6$ by $0, \dots, 5$, and write $e_0, \dots, e_5$ for the standard coordinate vectors. For a square Hadamard matrix, $H^*H = 6I$ also implies $HH^* = 6I$. The equivalence relation and normalized character g_H are those of Section 1. Every Hadamard matrix can be *dephased*, meaning that its first row and first column are all ones, by multiplying rows and columns by elements of $\mathbb{T}$.

A *charge* is a vector $a \in \mathbb{Z}^6$ with $\sum_i a_i = 0$. Its order and character are

$$\text{ord}(a) = \sum_{a_i > 0} a_i, \quad x^a = \prod_{i=0}^5 x_i^{a_i} \quad (x \in \mathbb{T}^6).$$

For the column characters in (1), put

$$D = \{e_i - e_j : 0 \leq i, j \leq 5, i \neq j\}.$$

We retain $\alpha = (1, 1, 1, -1, -1, -1)$ from the introduction. In particular, $|g_H(a)| \leq 1$, $g_H(0) = 1$, and $g_H(-a) = \overline{g_H(a)}$. A permutation p acts by $(P_p a)_i = a_{p(i)}$. When we write a permuted charge, all coordinates are permuted together; we also abbreviate $P_p a$ as pa .

Balanced charges make g_H invariant under column phases. Multiplying row i by u_i multiplies $g_H(a)$ by u^a . Thus products whose total charge is zero are unchanged by row phases as well. These elementary observations will justify dephasing and the invariant moments below.

3 Injective moments and a dichotomy for one Hadamard matrix

For a matrix H and a row permutation π , write $H^\pi(i, k) = H(\pi(i), k)$. The following dichotomy is the single-matrix step in the proof of Theorem 1.1.

Corollary 3.1 (Dichotomy). *For every 6×6 Hadamard matrix H , at least one of the following conditions holds:*

- (M) $g_{H^\pi}(\alpha) = 0$ for every row permutation π ;
- (C) the entrywise square $H^{\circ 2}$ is a Hadamard matrix.

Condition (M) is the Fourier condition in Matolcsi, Ruzsa, and Weiner's conjecture [15, Conjecture 2.3]; Section 4 will classify condition (C). We prove the dichotomy from an exact identity between averages of nonnegative quantities, using only orthogonality and unimodularity of one matrix.

3.1 Injective moments

Fix a 6×6 Hadamard matrix H . An *exponent matrix* is an integer matrix $A = (A_{uv})$ whose row sums and column sums are all zero. For A of size $p \times q$, with $p, q \leq 6$, define

$$m_H(A) = \mathbb{E} \operatorname{Re} \left[\prod_{u=0}^{p-1} \prod_{v=0}^{q-1} K(\rho(u), \kappa(v))^{A_{uv}} \right]. \quad (3)$$

Here K is H or $H^\top$ with probability $1/2$ each, $\rho : \{0, \dots, p-1\} \rightarrow \{0, \dots, 5\}$ and $\kappa : \{0, \dots, q-1\} \rightarrow \{0, \dots, 5\}$ are uniformly chosen injections, and these three choices are independent. Negative exponents are well defined because every matrix entry has modulus one. We set $m_H(\emptyset) = 1$ for the empty exponent matrix. In particular,

$$|m_H(A)| \leq 1. \quad (4)$$

Permuting the rows or columns of A leaves this expectation unchanged. Replacing A by $-A$ conjugates the monomial and therefore leaves its real part unchanged. Also $m_H(A^\top) = m_H(A)$: transposition exchanges the two injections and interchanges the equally weighted choices of K . Finally, a zero row or column can be deleted. Indeed, its factor is one, and deleting a coordinate of a uniformly chosen injection leaves a uniformly

chosen injection on the remaining coordinates. These facts justify all the identifications of exponent matrices used below.

For a balanced integer row a , its order is $\text{ord}(a) = \sum_{j: a_j > 0} a_j$, consistently with the notation for charges. The following relation eliminates any row of order one.

Lemma 3.2 (Contraction of an order-one row). *Let A be a balanced exponent matrix with p rows and $q \leq 6$ columns, and suppose row i has order one. For $\ell \neq i$, let $A_{i \rightarrow \ell}$ be obtained by adding row i to row ℓ and then deleting row i . If $p \leq 6$, then*

$$(7 - p)m_H(A) = - \sum_{\substack{0 \leq \ell < p \\ \ell \neq i}} m_H(A_{i \rightarrow \ell}). \quad (5)$$

If $p = 7$, the valid relation is instead

$$\sum_{\substack{0 \leq \ell < 7 \\ \ell \neq i}} m_H(A_{i \rightarrow \ell}) = 0. \quad (6)$$

Every moment in (6) has at most six rows; no moment with seven injectively sampled rows is defined or used. The analogous relations hold for columns by transposition.

Proof. Since row i is balanced and has order one, it has a $+1$ in one column v_+ , a -1 in another column v_- , and zeros elsewhere. Fix K , the column injection κ , and an injective assignment of the other $p - 1$ rows. Write M for the product of the factors contributed by those rows. The columns $\kappa(v_+)$ and $\kappa(v_-)$ are distinct, so column orthogonality gives

$$M \sum_{t=0}^5 K(t, \kappa(v_+)) \overline{K(t, \kappa(v_-))} = 0.$$

Both choices of K are Hadamard matrices: for a square matrix, column orthogonality also gives row orthogonality.

When $p \leq 6$, there are $7 - p$ row labels not yet assigned. The sum over those labels is $7 - p$ times the conditional expectation obtained by assigning row i uniformly to an unused label. Averaging the already fixed assignments produces $(7 - p)m_H(A)$. For each occupied label, say that of row ℓ , the corresponding monomial adds the exponents of row i to those of row ℓ . Its average is $m_H(A_{i \rightarrow \ell})$. Taking real parts and averaging also over K and κ proves (5). If a contraction creates a zero row or column, deletion is justified by the marginal argument preceding the lemma.

When $p = 7$, the six other rows occupy all six available labels. There is no unused-label contribution. The same orthogonality equation is therefore precisely the sum of the six contractions in (6). $\square$

Repeatedly applying Lemma 3.2 to rows or columns of order one expresses any moment with both dimensions at most six as a rational linear combination of the empty moment and moments whose nonzero rows and columns all have order at least two. Each contraction reduces the number of rows or columns, so this procedure terminates. Choosing representatives under the symmetries above introduces no additional mathematical relation.

3.2 From products of characters to injective moments

For a permutation π of $\{0, \dots, 5\}$, write $K^\pi(i, k) = K(\pi(i), k)$. We use the notation

$$\langle F(\tilde{H}) \rangle_H = \frac{1}{2 \cdot 6!} \sum_{K \in (H, H^T)} \sum_{\pi \in S_6} F(K^\pi) \quad (7)$$

for the average over the two orientations and all row permutations. The ordered pair $(H, H^\top)$ in this formula is counted with multiplicity even when its entries coincide. Column permutations do not change $g_{\tilde{H}}$ and need not be included.

Let $(6)_k = 6 \cdot 5 \cdots (6 - k + 1)$, with $(6)_0 = 1$.

Lemma 3.3 (Equality-pattern expansion). *Let $a_1, \dots, a_n \in \mathbb{Z}^6$ be balanced charges satisfying $\sum_{j=1}^n a_j = 0$, and let A be the matrix with these columns. For a set partition $\mathcal{S}$ of $\{1, \dots, n\}$, form $A_{\mathcal{S}}$ by replacing the columns in each block $S \in \mathcal{S}$ by their sum $\sum_{j \in S} a_j$. Then*

$$\left\langle \operatorname{Re} \prod_{j=1}^n g_{\tilde{H}}(a_j) \right\rangle_H = \sum_{\substack{\mathcal{S} \text{ a set partition of } \{1, \dots, n\} \\ |\mathcal{S}| \leq 6}} \frac{(6)^{|\mathcal{S}|}}{6^n} m_H(A_{\mathcal{S}}). \quad (8)$$

The ordering of the blocks of $\mathcal{S}$ is immaterial.

Proof. Expanding the product of characters samples n columns independently and uniformly from the six columns of $\tilde{H}$. Group the summands by their equality pattern: two positions belong to the same block precisely when their column labels coincide. For a specified partition with k blocks, the distinct block labels have $(6)_k$ possible assignments, each with probability 6^{-n} . Conditional on this pattern, the block labels form a uniform injection. Multiplying the factors with the same column label adds their exponents, giving $A_{\mathcal{S}}$. Averaging over the row permutation, the orientation, and the real part gives exactly its injective moment. Partitions with more than six blocks have no assignments and contribute nothing. $\square$

The hypotheses on the charges ensure that every matrix in (8) has both row and column sums zero. Moreover, $g_H(a) = g_H(-a)$, so absolute squares of products are covered by the lemma. A zero charge may be omitted before applying the expansion, since its factor is $g_H(0) = 1$; equivalently, integrating out its independently sampled column does not change the distribution of the remaining columns.

For example, if a is a balanced charge and $[a, -a]$ denotes the two-column exponent matrix, the two possible equality patterns give

$$\langle |g_{\tilde{H}}(a)|^2 \rangle_H = \frac{1}{6} + \frac{5}{6} m_H([a, -a]).$$

The first term comes from equal sampled columns, where all exponents cancel; the second comes from two distinct columns. The certificate below uses the same expansion for products with up to four factors.

3.3 An exact identity and its consequence

The auxiliary expression below is chosen so that orthogonality yields a sum of nonnegative terms equal to zero. The resulting vanishing products will force the dichotomy.

Retain $\alpha = (1, 1, 1, -1, -1, -1)$ and put

$$\begin{aligned} w &= (-3, -2, 0, 1, 2, 2), \\ a_1 &= (-1, -1, 0, 0, 1, 1), & a_2 &= (-2, 0, 0, 0, 1, 1), \\ a_3 &= (-1, -1, 0, 1, 0, 1), & a_4 &= (-1, -1, 0, 1, 1, 0). \end{aligned}$$

Define

$$L_H = -g_H(w) + \sum_{j=1}^4 c_j g_H(a_j) g_H(w - a_j), \quad (c_1, c_2, c_3, c_4) = (2, 1, 1, 1), \quad (9)$$

and the two charges

$$q_{\text{cross}} = (2, 0, 0, -2, 0, 0), \quad q_{\text{same}} = (2, -2, 0, 0, 0, 0).$$

Every term defining L_H has total charge w . The nonzero entries of q_{cross} lie in opposite sign triples of α , whereas those of q_{same} lie in the same triple. These two choices will supply the squared-row inner products in the proof of the dichotomy.

Proposition 3.4 (Single-matrix certificate). *For every 6×6 Hadamard matrix H and for each $q \in \{q_{\text{cross}}, q_{\text{same}}\}$,*

$$3\langle |L_{\tilde{H}}|^2 \rangle_H + \langle |g_{\tilde{H}}(\alpha)g_{\tilde{H}}(q)|^2 \rangle_H = 0. \quad (10)$$

Proof. Expand the two absolute squares and apply Lemma 3.3 to each resulting product. Every product has total charge zero, at most four factors, and sum of the orders of its charges equal to ten. Consequently the resulting expression is a rational linear combination of injective moments. Apply Lemma 3.2 until all remaining nonzero row and column orders are at least two, and choose representatives under row and column permutations, negation, and transposition.

The relations used to reduce this expression further are (6), generated from exponent matrices with seven rows, two through four columns, positive row orders, column orders at least two, and total order at most eleven, with a chosen row of order one. The total order of an exponent matrix is the sum of its row orders, equivalently the sum of its column orders. The right sides of these relations are reduced by the same contraction procedure. Thus all relations used are consequences of Lemma 3.2 and the stated symmetries.

Section 7 verifies in exact rational arithmetic that substitution of these relations makes every coefficient of the expanded left side of (10) zero, for each of the two specified charges q . In particular, its elimination rules hold for every moment vector satisfying those relations, whether or not there are further relations between the moments. The complete finite computation, including the two coefficient comparisons, is included in Appendix A. The resulting identity therefore holds for the moment vector of every Hadamard matrix H , which proves the proposition. $\square$

Proof of Corollary 3.1. The average in (10) is finite, and each choice of orientation and row permutation has positive weight. Every summand is nonnegative. Proposition 3.4 therefore implies, separately for every ordering of the rows of H ,

$$g_{H^\pi}(\alpha)g_{H^\pi}(q_{\text{cross}}) = 0, \quad g_{H^\pi}(\alpha)g_{H^\pi}(q_{\text{same}}) = 0.$$

Suppose (M) fails. The ordering witnessing its failure divides the row indices into two triples I_+ and I_- on which the charge is $+1$ and -1 , respectively. Permutations within either triple do not change this nonzero character. By placing any chosen $i \in I_+$ and $j \in I_-$ in positions 0 and 3, the first product identity gives $g_H(2(e_i - e_j)) = 0$. By placing any two distinct indices of I_+ in positions 0 and 1, the second product identity gives the same vanishing for that pair. Interchanging the two triples conjugates the nonzero character, since $g_H(-a) = \overline{g_H(a)}$. The same argument therefore treats pairs within I_- and all remaining ordered cross pairs. Hence

$$g_H(2(e_i - e_j)) = 0 \quad \text{for every } i \neq j.$$

For distinct rows i, j , this says

$$\sum_{k=0}^5 H(i, k)^2 \overline{H(j, k)^2} = 6g_H(2(e_i - e_j)) = 0.$$

Every row of $H^{\circ 2}$ has squared norm six. Thus $H^{\circ 2}(H^{\circ 2})^* = 6I$, which for a square matrix also implies $(H^{\circ 2})^*H^{\circ 2} = 6I$. Its entries are unimodular, so (C) holds. $\square$

4 The cubic alternative

We now classify the second alternative of Corollary 3.1 and complete the proof of Theorem 1.1. Throughout this section, put $\omega = \exp(2\pi i/3)$. Matrices satisfying the two Hadamard conditions are called S-Hadamard matrices [12, Definition 2.1]. The following proposition proves the order-six classification conjectured by Phangara [21, Conjecture 3.1.6]. Its representative is the matrix in Tao's construction [23, p. 254].

Proposition 4.1 (Classification of the cubic alternative). *Suppose that H and its entrywise square $H^{\circ 2}$ are Hadamard matrices of order six. By multiplying rows and columns by unit complex numbers and permuting rows and columns, one can transform H into the matrix T in (2). In particular, every entry of the transformed matrix is a cube root of unity.*

Proof. Let $v = (v_0, \dots, v_5)$ be the entrywise ratio of two distinct rows of H , and set $p_k(v) = \sum_i v_i^k$. The two Hadamard conditions give $p_1(v) = p_2(v) = 0$. If e_k denotes the k th elementary symmetric function of these six coordinates, Newton's identities give $e_1 = e_2 = 0$. Since $|v_i| = 1$,

$$e_{6-k} = e_6 \overline{e_k},$$

so $e_4 = e_5 = 0$ as well. Consequently

$$\prod_{i=0}^5 (z - v_i) = z^6 - e_3 z^3 + e_6.$$

Factoring this as a quadratic in z^3 shows that the multiset of coordinates of v is the union of two rotated cube-root triples. The triples may coincide. Thus the cubed coordinates have either one value six times or two distinct values three times each. The same conclusion holds for ratios of columns, because both matrices also have orthogonal columns.

Dephase H , making its initial row and column equal to one. This preserves both Hadamard conditions. Every row of $H^{\circ 3}$ is now either all ones or has the value 1 three times and a value $x \neq 1$ three times. For a nonconstant row, call the three positions carrying x its support; its support excludes column zero.

All nonconstant rows have the same support. Indeed, suppose two distinct supports S, U have respective values $x, y \neq 1$, and put $k = |S \cap U|$. As S, U are three-element subsets of five positions, $k \in \{1, 2\}$. The cubed row ratio has the following values and multiplicities:

value	1	x	y^{-1}	xy^{-1}
multiplicity	k	$3 - k$	$3 - k$	k .

Every multiplicity is positive, whereas this ratio can have at most two distinct cubed values. The three values $1, x, y^{-1}$ therefore force $x = y^{-1}$. Next $xy^{-1} = x^2$ must equal 1 or x , which forces $x = y = -1$. The two multiplicities would then be $2k$ and $6 - 2k$, namely 2 and 4 in some order, contradicting the required multiplicities 3 and 3. Applying the same argument to columns proves the corresponding assertion for their supports.

Suppose that $H^{\circ 3}$ is not all ones. Its common nonconstant-row support consists of three columns. Each of those columns has exactly three non-one entries, so exactly three rows are nonconstant. Comparing their values in any one of these columns shows that their non-one values coincide. After arranging these two classes of rows and columns, the cubed matrix therefore has the form

$$H^{\circ 3} = \begin{pmatrix} J_3 & J_3 \\ J_3 & xJ_3 \end{pmatrix}, \quad x \neq 1, \tag{11}$$

where J_3 is the all-ones matrix of order three.

Restrict each row of H to the three columns of the xJ_3 block. For rows from opposite classes, the ratio cubes to x or x^{-1} on these columns and to 1 on the other three. The two rotated triples in that ratio are therefore separated by this division of the columns. Their sums are zero, so the restrictions of rows from opposite classes are mutually orthogonal.

Each class spans a space of dimension at least two on these three columns. To see this, suppose that the three restrictions in one class span a line. Each has squared norm three, so the inner product of any two has modulus three. Orthogonality of the full rows forces their complementary restrictions to have inner product of modulus three as well. Equality in Cauchy–Schwarz makes these complementary restrictions proportional. The three full rows would then lie in a space of dimension at most two, contrary to their orthogonality. Thus the two mutually orthogonal restricted spans have dimensions at least two, which is impossible in $\mathbb{C}^3$. This excludes (11). Every entry of the dephased H is consequently a cube root of unity.

In each noninitial row and column, orthogonality to the initial one implies that each cube root occurs twice. There is therefore exactly one additional entry equal to one in each noninitial row and column. These entries form a matching and may be placed on the diagonal by a column permutation. Write the other entries as $\omega^{s_{ij}}$, where $s_{ij} \in \{-1, 1\}$ and $i, j \in \{1, \dots, 5\}$ are distinct. Each such row has two plus signs and two minus signs.

For two distinct noninitial rows i, j , their ratio contains each root twice. They agree at column zero, disagree at columns i, j , and hence agree at exactly one of the remaining three columns. There are thus two sign mismatches among those three columns, giving

$$\prod_{k \notin \{0, i, j\}} s_{ik} s_{jk} = 1.$$

The product of the four signs in each complete noninitial row is also one. It follows that $s_{ij} s_{ji} = 1$, so $s_{ij} = s_{ji}$. The plus signs define a simple undirected graph of degree two on five vertices. Its components are cycles of length at least three; therefore it is a single five-cycle. A simultaneous permutation of the noninitial rows and columns now gives (2); Figure 1 shows its two off-diagonal exponent classes. For completeness, the displayed T is itself Hadamard: every difference of distinct exponent rows contains each residue modulo three twice, as seen by separating an initial-row pair, an adjacent pair on the cycle, and a nonadjacent pair. Its entrywise square is $\bar{T}$ and is Hadamard as well. $\square$

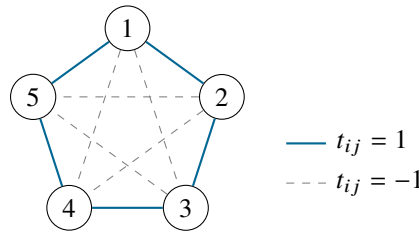


Figure 1: The five-cycle determining the noninitial off-diagonal entries of the cubic matrix T . Solid edges contribute ω and dashed edges contribute ω^{-1} . The diagonal and the initial row and column consist of ones.

The classification also settles the Fourier-vanishing conjecture of Matolcsi, Ruzsa, and Weiner [15, Conjecture 2.3]. Here equivalence means multiplication of rows and columns by unit complex numbers and permutation of rows and columns. The matrix T is equivalent to the spectral matrix S_6 displayed in [22, (93)–(94)]: with our zero-based indices, $T_{ij} = (S_6)_{p(i), p(j)}$ for $(p(0), \dots, p(5)) = (0, 1, 2, 5, 4, 3)$.

Proof of Theorem 1.1. If $g_H(\pi\alpha)$ were nonzero for some coordinate permutation π , Corollary 3.1 would imply that $H^{\circ 2}$ is Hadamard. Proposition 4.1 would then make H equivalent to T , a contradiction. $\square$

5 Complete families and the cubic obstruction

We turn to Theorem 1.2. The first step is to exclude the cubic alternative from a complete family. We begin with the character identities that record mutual unbiasedness and completeness.

5.1 Character identities of a complete family

Suppose that $B_1, \dots, B_7$ are mutually unbiased orthonormal bases. For $r \neq s$, define their transition matrix by

$$H_{rs}(i, k) = \sqrt{6} \langle b_{r,i}, b_{s,k} \rangle.$$

Unbiasedness and orthonormality imply that H_{rs} is Hadamard and that $H_{sr} = H_{rs}^*$. With a reference basis r fixed, we abbreviate H_{rs} and $g_{H_{rs}}$ to H_s and g_s .

Lemma 5.1. *For each fixed reference basis the following identities hold. First,*

$$g_s(d) = 0 \quad (d \in D), \quad \sum_{s \neq r} g_s(a) = 0 \quad (\text{ord}(a) = 2). \quad (12)$$

For arbitrary charges a, b and each coordinate i ,

$$\sum_{j=0}^5 g_s(a + e_i - e_j) g_s(b - e_i + e_j) = g_s(a + b). \quad (13)$$

The same identity holds with both shifts reversed. For distinct external bases $s, t \neq r$,

$$\sum_{d \in D} g_s(a + d) g_t(b - d) = 0. \quad (14)$$

The complete-set projector identity used in the proof is the second-moment identity for a complex projective 2-design; see Zauner [26, Satz 2.19] and Klappenecker–Rötteler [8, Theorem 3]. We give the projection argument in our normalization. The character identities are part of the Fourier approach [13, 15].

Proof. The first assertion in (12) is row orthogonality. For the second, let $P_{s,k} = b_{s,k} b_{s,k}^*$, and let $\mathcal{D}_s^0$ be the complex space of traceless matrices diagonal in B_s . If $s \neq t$, then

$$\text{Tr}((P_{s,k} - I/6)(P_{t,l} - I/6)) = 0.$$

The spaces $\mathcal{D}_s^0$ are therefore pairwise orthogonal for the Hilbert–Schmidt inner product. Each has dimension five, and their dimensions sum to $35 = 6^2 - 1$, the dimension of the traceless matrix space. Orthogonal projection consequently gives

$$\text{Tr}(XY) = \sum_{s=1}^7 \sum_{k=0}^5 \text{Tr}(X P_{s,k}) \text{Tr}(Y P_{s,k}) \quad (\text{Tr } X = \text{Tr } Y = 0). \quad (15)$$

This is a complex bilinear identity: it follows by pairing the complex-linear projection formula for X with Y .

Any order-two charge is a sum $d_1 + d_2$ of two members of D . For $s \neq r$, write h_k for the k th column of H_s . In basis B_r , choose off-diagonal matrix units X, Y whose traces against $P_{s,k}$, for $s \neq r$, are respectively $h_k^{d_1}/6$ and $h_k^{d_2}/6$. For example, $d_1 = e_i - e_j$ corresponds to $X = E_{ji}$. Since $d_1 + d_2$ has order two, $d_2 \neq -d_1$, so $\text{Tr}(XY) = 0$. The reference basis contributes zero to (15), and each external basis contributes $g_s(d_1 + d_2)/6$. This proves the second identity in (12).

For (13), expand the two character sums and write x, y for their sampled columns. The left-hand side is

$$\frac{1}{36} \sum_{x,y} x^a y^b \frac{x_i}{y_i} \sum_{j=0}^5 \frac{y_j}{x_j}.$$

The inner sum is 6 when $x = y$ and zero otherwise, by column orthogonality. The result is $g_s(a + b)$. Conjugating the inner orthogonality relation proves the version with reversed shifts.

Finally, if x is a column of H_s and y a column of H_t , their inner-product squared modulus is six. Hence

$$\sum_{d \in D} (x/y)^d = \left| \sum_{i=0}^5 \frac{x_i}{y_i} \right|^2 - 6 = 0.$$

Expanding the products of g_s and g_t proves (14). $\square$

The identities for one matrix hold for every Hadamard matrix. The order-two sum in (12) uses the completeness of the seven-basis set. The later mixed identities retain this completeness information through averaging over the six external bases.

5.2 The cubic-pair obstruction

Brierley and Weigert proved the stronger fact that the standard basis and the basis defined by $T/\sqrt{6}$ cannot be extended by a third mutually unbiased basis [2, Section 4.1 and Table 1]. The following proposition gives an exact Fourier derivation of the complete-family exclusion needed here.

Proposition 5.2. *In a set of seven mutually unbiased bases in $\mathbb{C}^6$, no transition Hadamard matrix H can have $H^{\circ 2}$ Hadamard.*

To prove the proposition, suppose that such a pair occurs. By Proposition 4.1, rephasing and reordering its basis vectors allows us to take its transition matrix to be T . These operations preserve every mutual unbiasedness condition. Use the first basis of this pair as reference, and call the other five transition matrices $K_1, \dots, K_5$. Define

$$f(b) = \frac{1}{5} \sum_{s=1}^5 g_{K_s}(b), \quad G(b) = g_T(b).$$

Lemma 5.1 gives $f(b) = G(b) = 0$ at order one, and Equation (12) gives

$$f(b) = -\frac{1}{5}G(b) \quad (\text{ord}(b) = 2). \quad (16)$$

Equation (14), applied with its second charge equal to $-c$ and then averaged over $K_1, \dots, K_5$, gives

$$\sum_{d \in D} f(a + d) \overline{G(c + d)} = 0 \quad \text{for all balanced } a, c. \quad (17)$$

Let

$$S(b) = \frac{1}{6!} \sum_{\pi \in S_6} \text{Re}(f(\pi b) \overline{G(\pi b)}). \quad (18)$$

Because $f(-b) = \overline{f(b)}$ and likewise for G , this average also equals the average over the distinct elements of the signed permutation orbit of b .

Every entry of T has cube one, so $G(3d) = 1$ for $d \in D$. The signed permutation orbit of $3(e_5 - e_0)$ is precisely $\{3d : d \in D\}$. If z is a uniformly sampled column of a uniformly sampled matrix among $K_1, \dots, K_5$, then $\mathbb{E}z^b = f(b)$. Consequently

$$0 \leq \mathbb{E} \left| \sum_{i=0}^5 z_i^3 \right|^2 = 6 + \sum_{d \in D} f(3d) = 6 + 30S(3(e_5 - e_0)). \quad (19)$$

Thus this orbit average must be at least $-1/5$. We will use (17) to compute it and contradict this lower bound. The next lemma concerns T alone; it lets us calculate the coefficients of those identities one permutation orbit at a time.

5.3 Symmetries of the cubic matrix

The row-permutation symmetry below also follows from the semilinear automorphism group determined by Gillespie, Ó Catháin, and Praeger [5, Proposition 2]. We give a direct proof using the five-cycle classification.

Lemma 5.3. *Every permutation of the rows of T can be completed to a symmetry of T by column permutations, row and column phase factors, and, if necessary, complex conjugation. Consequently $|G(b)|$ is constant on each orbit of balanced charges under permutations and negation.*

Proof. A symmetry operation consists of row and column permutations, a choice of whether to conjugate, and row and column phase factors that send T to itself. Normalize the phase factor of output row zero to one. The all-one output row and column then determine all the other phase factors, removing the freedom to multiply the row factors by a common phase and the column factors by its inverse.

Choose an initial row and column in 36 ways, choose whether to conjugate, and dephase at the chosen row and column. Proposition 4.1 applies again. After matching the additional ones on the diagonal, there are exactly 10 orders of the remaining rows that, together with the matching column orders, make the plus graph the specified five-cycle: choose its initial vertex and its direction. This constructs $36 \cdot 2 \cdot 10 = 720$ symmetry operations. Their row and column permutations recover the chosen initial row and column as the source indices sent to position zero; together with the conjugation choice, their remaining orders recover the chosen cycle isomorphism. Thus these are distinct normalized operations.

We show that their induced row permutations are distinct. Work with exponents modulo three, and write $t_j = (t_{ij})_{i=0}^5$ for column j of the exponent matrix. A symmetry fixing the row order acts on the set $X = \{t_0, \dots, t_5\}$ as

$$X \mapsto sX + u, \quad s \in \{1, -1\}.$$

Indeed, because row zero has only zero exponents, the column phase factors are common and may be absorbed into the row factors. The remaining row factors have exponents modulo three, and the image of the zero column shows that $u \in X$.

For $j \neq 0$, column t_j has exactly one zero in its noninitial positions, at position j . Hence $-t_j \notin X$: its zero would force it to equal t_j , which it does not. A nonzero translation preserving X would contain both u and $2u = -u$, and is therefore impossible. Negation with $u = 0$ is impossible for the same reason.

Finally suppose $s = -1$ and $u = t_x \neq 0$. Choose a neighbor y of x on the five-cycle, and let z be their common non-neighbor. Directly from (2), $t_x - t_y$ has its unique noninitial zero at position z : at the other two positions outside $\{x, y, z\}$, exactly one of x, y is adjacent to that vertex, so the differences are ± 2 , hence nonzero modulo three. Were $t_x - t_y$ a column, it would have to be t_z . But at positions x, y the former has entries $-1, 1$, while t_z has entries $-1, -1$, a contradiction modulo three. Thus the only symmetry fixing the row order is the identity, after the irrelevant common phase normalization. Two of the 720 operations with the same row permutation would have such a quotient, so their row permutations are all distinct. They exhaust the 720 possible row permutations.

For a balanced charge, column phases cancel in the defining character, while row phases multiply it by a unit complex number. Conjugation conjugates the character. These observations prove the last assertion, including negation since $G(-b) = \overline{G(b)}$. $\square$

5.4 Coefficients of the cross identities

Let

$$a = (-1, -1, 0, 0, 1, 1), \quad c = (0, 0, 0, 0, -1, 1),$$

and let O be the set of distinct simultaneous permutations (A, C) of (a, c) . Apply (17) to every $(A, C) \in O$, multiply by $\overline{G(A - C)}$, sum, and take real parts. To record the coefficients explicitly, set

$$Q(b) = \sum_{\substack{(A,C) \in O \\ b-A \in D}} G(A - C)G(b - A + C).$$

The resulting equality is

$$\sum_b \operatorname{Re}(f(b)\overline{Q(b)}) = 0,$$

where the sum is finite. For a complex number z , write $z^{[1]} = z$ and $z^{[-1]} = \bar{z}$. Conjugacy of f permits symmetrizing the coefficient as $R(b) = (Q(b) + \overline{Q(-b)})/2$: replacing b by $-b$ in the finite sum above and averaging leaves it unchanged. Explicitly,

$$R(b) = \frac{1}{2} \sum_{\substack{(A,C) \in O, \epsilon \in \{-1,1\} \\ \epsilon b - A \in D}} [G(A - C)G(\epsilon b - A + C)]^{[\epsilon]}. \quad (20)$$

We have therefore obtained the weighted cross identity

$$\sum_b \operatorname{Re}(f(b)\overline{R(b)}) = 0. \quad (21)$$

Its coefficients depend only on T . We now calculate them.

The following table lists representatives w_j of every balanced charge of order at most three, up to permutation and negation. The integer triples in its last two columns satisfy

$$6G(w_j) = \sum_{m=0}^2 N_m \omega^m, \quad 72R(w_j) = \sum_{m=0}^2 U_m \omega^m. \quad (22)$$

Here is an explicit summation rule reproducing Table 1. Put

$$v(b, k) = \sum_{i=0}^5 b_i t_{ik} \pmod{3},$$

with residues in $\{0, 1, 2\}$. For each input $b = w_j$, initialize two separate arrays N, U of three zero integers and apply these loops:

```
for k = 0, ..., 5:
  N[v(b,k)] += 1
for p,q = 0, ..., 5 with p != q:
  for 0 <= h < i <= 5 with {h,i} disjoint from {p,q}:
```

j	w_j	(N_0, N_1, N_2)	(U_0, U_1, U_2)
0	$(-3, 0, 0, 0, 0, 3)$	$(6, 0, 0)$	$(0, 0, 0)$
1	$(-3, 0, 0, 0, 1, 2)$	$(2, 2, 2)$	$(0, 0, 0)$
2	$(-3, 0, 0, 1, 1, 1)$	$(3, 0, 3)$	$(0, 0, 0)$
3	$(-2, -1, 0, 0, 1, 2)$	$(4, 1, 1)$	$(60, 42, 42)$
4	$(-2, -1, 0, 1, 1, 1)$	$(2, 2, 2)$	$(144, 144, 144)$
5	$(-2, 0, 0, 0, 0, 2)$	$(2, 2, 2)$	$(0, 0, 0)$
6	$(-2, 0, 0, 0, 1, 1)$	$(3, 3, 0)$	$(180, 180, 72)$
7	$(-1, -1, -1, 1, 1, 1)$	$(3, 3, 0)$	$(540, 540, 216)$
8	$(-1, -1, 0, 0, 1, 1)$	$(4, 1, 1)$	$(480, 336, 336)$
9	$(-1, 0, 0, 0, 0, 1)$	$(2, 2, 2)$	$(576, 576, 576)$
10	$(0, 0, 0, 0, 0, 0)$	$(6, 0, 0)$	$(0, 0, 0)$

Table 1: Exact cube-root counts for G and R .

```

A = e_p + e_q - e_h - e_i
u = 2*e_p - e_h - e_i
for s in {-1,1} with s*b - A in D:
    for k,l = 0,...,5:
        U[(s*(v(u,k) + v(s*b-u,l))) mod 3] += 1

```

Indeed, the $30\binom{4}{2} = 180$ pairs in $\mathcal{O}$ are uniquely parametrized by

$$A = e_p + e_q - e_h - e_i, \quad C = e_q - e_p,$$

with precisely the conditions used in the loops; thus $u = A - C$. Each product of two G terms contributes 36 cube roots with coefficient $1/36$, and the factor $1/2$ in (20) accounts for the denominator 72 in (22).

For example, row 3 gives $G(w_3) = (4 + \omega + \omega^2)/6 = 1/2$ and $R(w_3) = (60 + 42\omega + 42\omega^2)/72 = 1/4$. Using $1 + \omega + \omega^2 = 0$ in every row gives

$$R(b) = \lambda_j G(b), \quad \lambda_3 = \frac{1}{2}, \quad \lambda_6 = 3, \quad \lambda_7 = 9, \quad \lambda_8 = 4, \quad (23)$$

on the signed permutation orbit of w_j , and $R(b) = 0$ on all other orbits. To justify extending the representative calculation, observe that every factor in (20) has a balanced charge. Its column phases therefore cancel. The two charges in each product add to ϵb , so after the prescribed conjugation its row phase factor is exactly the factor for $G(b)$. Row permutations preserve the full set $\mathcal{O}$, and conjugation conjugates both R and G . Lemma 5.3 therefore gives the asserted covariance on every orbit; also $R(-b) = \overline{R(b)}$. No other orders can occur in the support of R , since $\epsilon b = A + d$ with $\text{ord}(A) = 2$ and $d \in D$ implies $\text{ord}(b) \leq 3$.

5.5 The contradiction

Completion of the proof of Proposition 5.2. Write $S_j = S(w_j)$ for the representatives in Table 1. The orbit sizes for $j = 3, 6, 7, 8$ are respectively 360, 120, 20, 90. By (23), their coefficients in (21) are 180, 360, 180, 360. Dividing by 180 proves

$$S_3 + 2S_6 + S_7 + 2S_8 = 0. \quad (24)$$

Table 1 and Lemma 5.3 imply

$$S_1 = S_4 = S_5 = S_9 = 0.$$

At types 6 and 8 the squared modulus of G is $1/4$, so (16) gives

$$S_6 = S_8 = -\frac{1}{20}. \quad (25)$$

Next take $a = c = w_5$ in (17) and average over coordinate permutations. Do the same with $a = c = w_8$. Classifying the 30 charges $w_j + d$, $d \in D$, by their signed permutation orbits gives the following counts:

	0	1	3	4	6	7	8	9
$w_5 + D$	1	8	12	0	8	0	0	1
$w_8 + D$	0	0	4	8	4	2	8	4

These counts can be obtained simply by adding each of the 30 vectors $e_i - e_j$ to the displayed representatives and sorting the coordinates, allowing a simultaneous change of sign. Averaging over permutations turns each resulting charge into its orbit average in (18). Substituting the known values gives

$$S_0 + 12S_3 = \frac{2}{5}, \quad 4S_3 + 2S_7 = \frac{3}{5}. \quad (26)$$

Together, (24), (25), and (26) yield

$$S_3 = S_7 = \frac{1}{10}, \quad S_0 = -\frac{4}{5}.$$

Since $w_0 = 3(e_5 - e_0)$, equation (19) now gives

$$\mathbb{E} \left| \sum_{i=0}^5 z_i^3 \right|^2 = 6 + 30S_0 = -18,$$

contradicting nonnegativity. This proves the proposition. $\square$

6 Mixed moments and the remaining case

Suppose that a complete set of seven mutually unbiased bases exists and that its transition matrices satisfy

$$g_{H_{rs}}(P\alpha) = 0 \quad (r \neq s, P \text{ a coordinate permutation}). \quad (27)$$

We seek a nonnegative averaged quadratic form whose expansion in bounded moments has a strictly negative upper bound. Corollary 3.1 and Proposition 5.2 will then exclude a complete set.

6.1 Sampling and positive forms

Choose a reference basis index r uniformly from the seven indices. Given r , choose independently a uniform permutation σ of its six coordinates and a uniform bijection π from $\{0, \dots, 5\}$ to the six other basis indices. In this section write

$$H_s(i, k) = \sqrt{6} \langle b_{r, \sigma(i)}, b_{\pi(s), k} \rangle, \quad g_s = g_{H_s}.$$

The column order within each basis may remain fixed: each g_s already averages over all six columns. For labels $\mathbf{s} = (s_1, \dots, s_n)$ and balanced charges $\mathbf{a} = (a_1, \dots, a_n)$ with $\sum_j a_j = 0$, define

$$\mathcal{M}(\mathbf{s}; \mathbf{a}) = \mathbb{E} \operatorname{Re} \left(\prod_{j=1}^n g_{s_j}(a_j) \right). \quad (28)$$

Equal labels always denote the same matrix. The marginal assignment of any l distinct labels is a uniform injection into the six nonreference bases. Thus the definition does not depend on which unused labels are included in the sampled bijection. The empty product has moment 1.

We also use the ordered-pair average of the injective moments of Section 3:

$$m(A) = \frac{1}{42} \sum_{r \neq s} m_{H_{rs}}(A), \quad H_{rs}(i, k) = \sqrt{6} \langle b_{r,i}, b_{s,k} \rangle. \quad (29)$$

Here m_H is the real injective moment with the transpose averaging specified there. All these actual moments obey

$$|m(A)| \leq 1, \quad |\mathcal{M}(\mathbf{s}; \mathbf{a})| \leq 1. \quad (30)$$

For the first inequality the integrand has modulus one. For the second, each $g_s(a)$ is an average of six numbers of modulus one.

The next construction expresses the real parts of Gram entries as linear combinations of three-factor mixed moments. The third factor comes from one column shared by all the functions.

In addition to the preceding sampling, choose a uniform column z of H_0 . For every balanced charge a , define functions on this probability space by

$$F_0(a) = g_0(a)z^{-a}, \quad F_1(a) = \left(\frac{1}{5} \sum_{s=1}^5 g_s(a) \right) z^{-a}. \quad (31)$$

The same sampled column z is used in every function $F_t(a)$. This use of moments and positive Gram forms follows the broader polynomial-optimization approach to MUBs [3, 6]; the shared-column functions above specify the particular forms used here.

In the formulas below, an index 1 on F denotes the average over five bases in (31); a label 1 in $\mathcal{M}$ denotes a representative of those bases after taking expectation.

Lemma 6.1 (Gram entries). *For balanced a, v , put $\mathbf{b} = (a, -v, v - a)$. If $(s, t) \in \{0, 1\}^2$ and $(s, t) \neq (1, 1)$, then*

$$\operatorname{Re} \mathbb{E}(F_s(a) \overline{F_t(v)}) = \mathcal{M}((s, t, 0); \mathbf{b}). \quad (32)$$

For the remaining case,

$$\operatorname{Re} \mathbb{E}(F_1(a) \overline{F_1(v)}) = \frac{1}{5} \mathcal{M}((1, 1, 0); \mathbf{b}) + \frac{4}{5} \mathcal{M}((1, 2, 0); \mathbf{b}). \quad (33)$$

Proof. Conditioning first on all the matrices and averaging over z gives $\mathbb{E}_z z^{v-a} = g_0(v-a)$. In a term with exactly one F_1 , the five summands have the same expectation by label exchangeability. In a term with two F_1 factors, expand the two sums independently. Of the 25 ordered choices of labels from $\{1, \dots, 5\}$, five have equal labels and twenty have distinct labels. Their marginal distributions give (33). This averaging removes the auxiliary column completely. A zero charge contributes $g_s(0) = 1$ and may be deleted; removing any unused label preserves the uniform injection law already described. $\square$

6.2 Reducing the Gram entries

The three-factor moments in Lemma 6.1 can be simplified using symmetry, character vanishing, and the identities of a complete family. When all factors refer to one matrix, the injective moments and equations from Section 3 apply as well.

Lemma 6.2 (Mixed-moment reductions). *Under (27), the following operations and identities are valid.*

- (i) *A common coordinate permutation of all charges, simultaneous negation of all charges, a simultaneous reordering of factors and their labels, and an arbitrary injective renaming of the used labels preserve (28). A zero charge factor may be omitted. Coordinates on which every charge is zero may also be omitted, with the remaining coordinates interpreted as a uniform injection into the six row positions.*
- (ii) *The moment vanishes if any charge has order one or is a coordinate permutation of α .*
- (iii) *Suppose that a_j has order two and its label s_j occurs only once. Let S be the set of distinct used labels and $l = |S|$. For each $t \in S \setminus \{s_j\}$, let $\mathbf{s}^{j \rightarrow t}$ be obtained from $\mathbf{s}$ by replacing s_j with t . Then*

$$\mathcal{M}(\mathbf{s}; \mathbf{a}) = -\frac{1}{7-l} \sum_{t \in S \setminus \{s_j\}} \mathcal{M}(\mathbf{s}^{j \rightarrow t}; \mathbf{a}). \quad (34)$$

- (iv) *If all labels are equal, the moment is a linear combination of the pair moments (29). More precisely, let A have columns $a_1, \dots, a_n$, after removing its zero rows. For a set partition $\mathcal{P}$ of $\{1, \dots, n\}$, let $A_{\mathcal{P}}$ have one column $\sum_{j \in B} a_j$ for each block $B \in \mathcal{P}$. Then*

$$\mathcal{M}((s, \dots, s); \mathbf{a}) = \sum_{\substack{\mathcal{P} \text{ a set partition} \\ |\mathcal{P}| \leq 6}} \frac{(6)_{|\mathcal{P}|}}{6^n} m(A_{\mathcal{P}}), \quad (35)$$

where $(6)_k = 6 \cdot 5 \cdots (6 - k + 1)$. Zero rows or columns in $A_{\mathcal{P}}$ may be deleted as in Section 3.

- (v) *For balanced a, b, c with $a + b + c = 0$,*

$$\sum_{j=0}^5 \mathcal{M}((s, s, u); (a + e_i - e_j, b - e_i + e_j, c)) = \mathcal{M}((s, u); (a + b, c)), \quad (36)$$

$$\sum_{d \in D} \mathcal{M}((s, t, u); (a + d, b - d, c)) = 0 \quad (s \neq t). \quad (37)$$

Equation (36) also holds with the signs of both shifts reversed. The third label u may coincide with another label.

Proof. Uniform sampling gives the coordinate and label symmetries in (i), while commutativity gives factor reordering. Simultaneous negation conjugates the whole product, leaving its real part unchanged. A zero charge gives $g_s(0) = 1$. Removing unused labels or coordinates leaves a uniform injection on those retained. Part (ii) follows from Hadamard orthogonality and (27). In particular, a permuted α has six nonzero coordinates; deleting zero coordinate rows cannot change whether a charge is of this type.

For (iii), fix r , the row permutation, and the bases assigned to the $l - 1$ other used labels. There are $6 - (l - 1) = 7 - l$ available bases for the singleton label. For this fixed row order, let $\tilde{g}_v$ denote the character average for the external basis v . Equation (12) gives

$$\sum_{\substack{v \neq r \\ v \text{ not already assigned}}} \tilde{g}_v(a_j) = - \sum_{t \in S \setminus \{s_j\}} g_t(a_j),$$

with the same row order throughout. Multiply by the other factors and average over the remaining choices. The marginal distribution of the $l - 1$ retained labels is exactly the distribution defining each moment on the right of (34).

For (iv), expand each g_s as its column average and group the resulting n column choices by their equality partition. A partition with k blocks has $(6)_k$ distinct assignments to its blocks, out of 6^n choices, giving the displayed coefficient. The retained rows are sampled injectively. It remains to justify the transpose averaging in $m(A)$. The marginal ordered pair of bases is uniform, and $H_{sr} = \overline{H_{rs}}^\top$. Exchanging r and s exchanges the row and column injections and sends an exponent matrix A to $-A^\top$. Negating all exponents conjugates the monomial, leaving its real part unchanged. Thus the ordered-pair average is already invariant under transposition. This proves (35), including the transpose convention in (29).

Finally, multiply (13) or (14) by $g_u(c)$ and take real expectations. These pointwise identities give (v). $\square$

In the finite calculation below, a mixed key consists of its factor labels and the matrix whose columns are its charges. The reductions in Lemma 6.2 apply to these keys. Single-label keys are expanded into the moments $m(A)$, and the pair relations from Section 3 may then be used. If a single-label expression vanishes by part (ii), its expansion also gives a valid homogeneous relation among the $m(A)$. These steps preserve the interpretation of every remaining symbol as an actual moment.

6.3 The six positive forms

Here are the finite collections of functions and integer weights used in the certificate. The exact data are the literal list named `data` in Appendix A, also supplied in the companion file `certificate/verify.py`. Each record has the form

$$(\lambda, \mathbf{ws}, e_\lambda, \mathbf{a}).$$

The list $\mathbf{ws} = ((t_0, v_0), \dots, (t_{m-1}, v_{m-1}))$ specifies m pairs with $t_i \in \{0, 1\}$ and balanced $v_i \in \mathbb{Z}^6$. The integer list $\mathbf{a}$ specifies a symmetric $m \times m$ matrix Z_λ by its lower triangle in row order: with zero-based list indices,

$$(Z_\lambda)_{ij} = (Z_\lambda)_{ji} = \mathbf{a}_{i(i+1)/2+j} \quad (0 \leq j \leq i < m). \quad (38)$$

Thus the data define the matrices and lists without any optimization or choice of numerical tolerances. Their sizes and scales are

λ	m	e_λ
(1, 1, 1, 2)	5	14
(1, 1, 3)	21	13
(1, 2, 2)	11	13
(1, 4)	27	14
(2, 3)	22	14
(5)	12	15

For each of these six lists of row lengths λ , place the positions $1, \dots, 5$, in order, in successive left-justified rows of lengths $\lambda_1, \lambda_2, \dots$. Let R_λ consist of permutations within these rows, and let C_λ consist of permutations within the resulting columns. Both groups act on $\{0, \dots, 5\}$ by fixing position 0. Our convention for their action on charges is

$$(P_p v)_i = v_{p(i)}.$$

These signed row and column permutation sums are the usual Young-symmetrizer construction. Symmetry reduction of positive Gram forms is developed generally in [4] and for MUB moment problems in [6]. Here the finite sums are given explicitly, so their use requires no representation-theoretic decomposition theorem. Define the vector Y_λ by

$$(Y_\lambda)_i = \sum_{\substack{p \in R_\lambda \\ q \in C_\lambda}} \text{sgn}(q) F_{t_i}(P_p P_q v_i) \quad (0 \leq i < m). \quad (39)$$

In particular, $(P_p P_q v)_i = v_{q(p(i))}$, which is the order implemented by $\text{perm}(\text{perm}(\mathbf{v}, \mathbf{q}), \mathbf{p})$ in the appendix. Formula (39) defines a finite linear combination of the functions (31) for every listed pair.

Proposition 6.3 (Mixed-moment certificate). *Each matrix Z_λ specified above is positive definite. Under the mixed-moment identities of Lemma 6.2, the forms (39) satisfy*

$$\begin{aligned} 5 \sum_{\lambda} 10^{15-e_\lambda} \mathbb{E}(Y_\lambda^* Z_\lambda Y_\lambda) &\leq \frac{-613302797399911 + 179721388988719}{6480} \\ &= -\frac{2007321335237}{30} < 0. \end{aligned} \quad (40)$$

Proof. The exact finite calculations establishing the matrix and coefficient claims are detailed in Section 7, with their full integer and rational arithmetic in Appendix A. We specify here the mathematical expansion to which those checks apply.

First expand each Y_λ by (39), and replace every real Gram entry using Lemma 6.1. Multiplication by five makes its weights either 5 or the pair (1, 4). Because Z_λ is real symmetric, summing its diagonal contributions and twice its lower off-diagonal contributions gives the real quantity $\mathbb{E}(Y_\lambda^* Z_\lambda Y_\lambda)$. The factor 10^{15-e_λ} is retained for each block.

Apply the mixed reductions and the homogeneous relations (36)–(37), using the finite selection of relations with total charge order at most 11 described in Section 7. That selection only restricts which valid relations are used. The exact substitutions give

$$5 \sum_{\lambda} 10^{15-e_\lambda} \mathbb{E}(Y_\lambda^* Z_\lambda Y_\lambda) = f + \sum_{\rho \in \mathcal{R}} \beta_\rho X_\rho, \quad (41)$$

where each X_ρ is a nonconstant symbol representing one of the original moments $m(A)$ or $\mathcal{M}(\mathbf{s}; \mathbf{a})$, and

$$f = -\frac{613302797399911}{6480}, \quad \sum_{\rho \in \mathcal{R}} |\beta_\rho| = \frac{179721388988719}{6480}. \quad (42)$$

The linear-algebra verification in Section 7 proves that these substitutions follow from the chosen equations. The bounds (30) applied to (41) now prove (40).

For each Z_λ , the exact Schur-complement calculation in Section 7 verifies positive pivots and hence positive definiteness. $\square$

Completion of the proof of Theorem 1.2. If seven mutually unbiased bases existed, Corollary 3.1 would give case (M) or case (C) for each transition matrix. Proposition 5.2 excludes case (C), so all pairs satisfy (27) and the preceding construction applies. For a complex vector $Y = x + iy$ and a real symmetric positive definite matrix Z ,

$$Y^* Z Y = x^T Z x + y^T Z y \geq 0.$$

Consequently the left side of (40) is nonnegative. Its strictly negative upper bound is a contradiction. This proves Theorem 1.2. $\square$

7 Exact verification of the two certificates

We now give the finite calculations establishing Propositions 3.4 and 6.3. The complete program and all its data are printed in Appendix A. It specifies the integer matrices, coordinate orders, and finite enumerations used below. All arithmetic that certifies an equality or inequality is integer or rational arithmetic. Modular arithmetic supplies candidate substitutions, whose validity is then checked exactly. Rational Gram certificates

and exact positivity checks are established methods; see Peyrl–Parrilo [20, Section 3]. Our input matrices are specified integers, and the proof below certifies the displayed identities directly from those data.

The calculation has three steps. First, expand each expression in named moments using the identities already proved. Next, use a finite set of true linear equations to substitute for some of these moments. Finally, check either that every coefficient vanishes, for the pair certificate, or that the constant coefficient plus the absolute sum of the remaining coefficients is negative, for the mixed certificate. We first justify the substitutions for arbitrary real moment values, then specify the equations and coefficient checks for each certificate.

7.1 Exact certification of linear substitutions

Both certificates use finite systems of linear equations in their moment coordinates. The same substitution procedure applies to either system. Include the constant moment as a coordinate, so that every moment equation is homogeneous in the enlarged coordinate vector. Clearing denominators turns the equations into integer rows. The lists called $\mathbf{I}$ in the program specify a fixed priority order on the coordinates. This choice affects the form of the answer and, in the mixed calculation, the size of its residual coefficient sum; it does not add any relation.

Lemma 7.1 (Exact substitutions). *Let $W \in \mathbb{Z}^{r \times N}$ consist of selected moment equations after their denominators have been cleared. Suppose a set J of r columns gives a square submatrix W_J invertible modulo a prime. Put $F = \{1, \dots, N\} \setminus J$. If a rational matrix $P \in \mathbb{Q}^{N \times N}$ satisfies*

$$P_{F,F} = I, \quad P_{*,J} = 0, \quad WP = 0,$$

then every real vector $\mathbf{m}$ satisfying $W\mathbf{m} = 0$ obeys $\mathbf{m} = P\mathbf{m}$. Thus the rows of P give valid substitutions for all coordinates in terms of the coordinates indexed by F .

Proof. Invertibility modulo the prime implies that the integer $\det W_J$ is nonzero. In the coordinate order J, F , write

$$W = (W_J \quad W_F), \quad P = \begin{pmatrix} 0 & P_{J,F} \\ 0 & I \end{pmatrix}.$$

The exact equality $WP = 0$ gives $P_{J,F} = -W_J^{-1}W_F$. Every solution of $W\mathbf{m} = 0$ consequently satisfies $\mathbf{m}_J = P_{J,F}\mathbf{m}_F$, while the free coordinates of $P\mathbf{m}$ are already $\mathbf{m}_F$. This proves the claim over $\mathbb{R}$, regardless of whether the moment values are rational. Equivalently, P has rank $N - r$, its image equals $\ker W$, and its identity rules make it the identity on that kernel. $\square$

Here is how `eliminate` verifies the hypotheses of the lemma. Its first modular elimination uses the prime 1000000007. The routine `echelon` retains an original integer equation whenever that equation supplies a new pivot. These retained equations form W . Each pivot is at the largest remaining coordinate of its row; normalized elimination therefore proves that W_J is invertible modulo the prime. Back substitution processes pivot coordinates in ascending order. When a smaller pivot is removed, its already reduced row contains only that pivot and free coordinates. The final rule for a pivot thus has no other pivot on its right side.

The program then works modulo the further moduli listed in the source, retaining matching pivot sets, combines coefficients by the Chinese remainder algorithm, and uses `rat` to propose rational rules. These steps preserve the absence of other pivot coordinates. All nonpivot coordinates receive explicit identity rules. Finally, for each retained integer equation the program substitutes every proposed rule, clears all denominators, and checks that every integer coefficient is zero. This is the exact check $WP = 0$ in Lemma 7.1.

Every actual moment vector satisfies the selected equations, so this check suffices even if additional generated equations have not been used. In particular, the modular pivot count is not asserted to be the full

rational rank of the original system. A prime that conceals an additional independent equation merely leaves that true equation unused. The rational reconstruction and its size threshold are ways to find a candidate P ; the exact check and the lemma establish its validity.

7.2 Pair equations and the vanishing identity

For an integer array A with zero row and column sums, put

$$\text{wt}(A) = \sum_i \text{ord}(A_{i,*}).$$

This also equals the sum of the column orders. Arrays name the real injective moments $m_H(A)$ of Section 3; the empty array names the constant moment 1. A formal linear expression in these moments is stored as a finite list of rational coefficients indexed by arrays. Coefficients at the same index are added, and zero coefficients are discarded. These are the operations `add` and `lin` in the program.

The routine `canonpair` selects a representative using precisely the symmetries of m_H : permutation of either axis, removal of zero rows and columns, global negation, and transposition. More explicitly, after removing zero axes it considers $A, -A, A^T, -A^T$, retains orientations with at least as many rows as columns, orders columns by their positive-part orders, and minimizes over the remaining column permutations after lexicographically sorting the rows. Each operation preserves the moment by its definition and Lemma 3.2. The degree orders and lexicographic choices serve only to select consistent names.

Define $U_0(A)$ recursively as a linear expression in representatives whose nonzero row and column orders are all at least two. If a row of A , or of A^T , has order one, apply Lemma 3.2 in that orientation:

$$U_0(A) = -\frac{1}{7-p} \sum_{\ell \neq i} U_0(A_{i \rightarrow \ell}), \quad p \leq 6. \quad (43)$$

Canonicalize each contracted array before continuing. If no such row exists on either axis, retain its moment as a single coordinate. The empty array is retained in the same way. Every contraction reduces the sum of the two axis lengths, so this recursion terminates. This is `evpair`; `expand_degree1` supplies the unsigned sum of contractions. The program checks that every evaluated array has both dimensions at most six and has zero sums on both axes.

The pair table is formed from formal seven-row equations. The exact enumeration restrictions are as follows:

- (i) Choose a total weight $k \in \{7, 8, 9, 10, 11\}$ and a column count q with $2 \leq q \leq \min(4, \lfloor k/2 \rfloor)$.
- (ii) Choose nondecreasing sequences of seven row orders $(r_0, \dots, r_6)$ and q column orders $(c_0, \dots, c_{q-1})$. Both sequences sum to k , and

$$1 \leq r_i \leq \lfloor k/2 \rfloor, \quad 2 \leq c_j \leq \lfloor k/2 \rfloor.$$

- (iii) Enumerate integer $7 \times q$ arrays with zero row and column sums and exactly these orders. Rows of equal order may be put in lexicographic order, and arrays with the same canonical representative may be skipped.

The routines `parts`, `comps`, and `allAB` implement this enumeration. A row is generated as the difference of disjointly supported nonnegative compositions of its prescribed order, within the remaining positive and negative column capacities. The capacities enforce the prescribed column orders. The first row has order one: seven positive integer row orders with sum at most eleven cannot all be at least two, and the orders are sorted.

For each retained array, the seven-row boundary case of Lemma 3.2 gives the equation

$$\sum_{\ell=1}^6 U_0(A_{0 \rightarrow \ell}) = 0. \quad (44)$$

Here $0, \dots, 6$ index the formal rows, not coordinates of a charge. Only the contracted arrays, which have at most six rows, are evaluated. The checks in `pairtable` verify the balance, number of formal rows, and order of the first row before forming this equation. Skipping an array can only omit an available true equation; the proof does not require a complete set of moment relations.

Apply Lemma 7.1 to these equations. Write $U(A)$ for the resulting expansion of $U_0(A)$ in the retained moment coordinates. For a product used here, with charge columns $a_1, \dots, a_n$ and $n \leq 4$, let A_S be the array obtained by adding the columns in each block of a set partition S of $\{1, \dots, n\}$. The final pair expansion is

$$\sum_{\substack{S \text{ a set partition} \\ |S| \leq 6}} \frac{(6)_{|S|}}{6^n} U(A_S). \quad (45)$$

This is exactly Lemma 3.3. The routine `partition` enumerates set partitions by numbering blocks in order of first occurrence, and `rep_pair` implements (45). The routine `apair` first forms the array of charge columns. It may remove a zero charge before the expansion, since that removes the factor $g_H(0) = 1$. Deletion of zero coordinate rows is also legitimate: a uniform injection on the retained positions has the required uniform marginal.

For the pair certificate, `check_pair` forms the displayed polynomial L_H of Proposition 3.4, with coefficients $-1, 2, 1, 1, 1$. It multiplies each monomial by the conjugate of every monomial, expanding $3|L_H|^2$ through (45). For each of the two stated charges q , it adds the expansion with charge columns $(\alpha, q, -\alpha, -q)$. After the exact pair substitutions, the result has no nonzero coefficient, for either q . This proves the identity in Proposition 3.4; all expressions used in the check have been identified with their actual averaged moment values above.

7.3 Mixed keys and the finite mixed table

We use the notation $M(\mathbf{s}; \mathbf{a})$ and $m(A)$ from Section 6. The latter is the average of $m_H(A)$ over the ordered basis pairs. A mixed key consists of a label tuple $\mathbf{s}$ and an array A whose columns are the charges. The array has at most six nonzero coordinate rows. Its rows and columns sum to zero. The routine `cmix` chooses representatives by simultaneous factor and label permutations, renaming labels by order of first occurrence, sorting the common coordinate rows, global negation, and removing zero factors and zero coordinate rows. These are exactly the symmetries justified in Lemma 6.2. The separate tag $(\emptyset, A)$ names the pair moment $m(A)$; the completely empty key names 1.

The routine `evmix` supplies the following recursive expansion of a key into named moment coordinates:

- (i) If a charge has order one or is a permutation of α , the mixed moment is zero.
- (ii) When at most one label remains, expand the product by (45), now using the averaged pair moments $m(A)$, and attach the pair tag to its coordinates. If this single-label expression was zero by the preceding rule, also record its pair expansion as a zero equation. These recorded equations are stored in `rows0`.
- (iii) Otherwise, if an order-two charge has a label occurring exactly once among the factors, let l be the number of labels in use and replace that label in turn by each of the other $l - 1$ labels. The original moment is $-1/(7 - l)$ times the sum of the resulting moments. Apply the same rules recursively to each term.

(iv) If none of these rules applies, retain the key as an original mixed-moment coordinate.

The zero rules and the singleton coefficient follow from Lemma 6.2. Each singleton replacement decreases the number of distinct labels. The single-label use of the pair table is legitimate because the real average over ordered pairs includes the required transpose symmetry, as proved in that lemma.

The finite enumeration of triples is given by `triples(11)`. For precision, it generates balanced charges $a, b, c \in \mathbb{Z}^6$ with

$$\begin{aligned} a + b + c &= 0, & 0 \leq \text{ord}(c) \leq \text{ord}(b) \leq \text{ord}(a) \leq 5, \\ \text{ord}(a) + \text{ord}(b) + \text{ord}(c) &\leq 11. \end{aligned}$$

Choose a up to a coordinate permutation and sign by taking the lesser of its sorted entries and the sorted entries of $-a$. Enumerate all balanced b of orders at most $\text{ord}(a)$, put $c = -a - b$, and retain those meeting the displayed restrictions. Sorting the six rows of the array (a, b, c) removes common-coordinate duplicates. Balanced vectors of a given order are generated by subtracting two disjointly supported nonnegative compositions of that order, which is the routine `vecs`.

For each generated array, `mixeq` first evaluates all five equality patterns on its three labels, registering any single-label zero equations. It then considers the distinct permutations of the three charge columns, choosing $a \leq b$ lexicographically to avoid duplicated first-two-column orderings. Write

$$a_{ij} = a + e_i - e_j, \quad b_{ij} = b - e_i + e_j \quad (0 \leq i, j \leq 5).$$

The following equations are included after applying `evmix` to every term:

- For different first two labels, sum the moments with charges (a_{ij}, b_{ij}, c) over all $i \neq j$, obtaining zero by (14). For nonzero c , use the label tuples $(0, 1, 2)$, $(0, 1, 0)$, and $(0, 1, 1)$; for $c = 0$, use $(0, 1, 0)$. Include this equation only if all 36 shifted triples have total order at most eleven.
- For equal first two labels, fix i , sum over all six j , and subtract the moment with charges $(-c, c)$ and labels $(0, u)$, by (13). Use $(0, 0, u)$ with $u = 0, 1$ when $c \neq 0$, and $u = 0$ when $c = 0$. Also use the reversed shifts $(a + e_j - e_i, b - e_j + e_i, c)$. Include each equation only if all its six shifted triples have total order at most eleven.

The third charge is unchanged in these sums because the equations have been multiplied by its factor. On the right side of the equal-label equation, $a + b = -c$, which explains both remaining charges and the labels selected by `s[-2:]` in the source.

Finally add the pair equations recorded in `rows0`. All coordinates encountered by these reductions are registered, including coordinates that cancel in a particular equation. With the specified coordinate order, `mixtable` applies Lemma 7.1 to the nonzero equations. A vanishing expression encountered later may still be replaced by zero: its vanishing is an independently proved rule, whether or not its pair expansion was used to strengthen this table. No completeness of the enumeration, and no vanishing of unlisted higher-order moments, is assumed.

The exact table dimensions are as follows. The witness column counts the independent equations used in Lemma 7.1.

Table	Nonzero equations	Coordinates	Witness rows
Pair	8871	909	897
Mixed	46041	3429	2767

7.4 Positive matrices and the negative bound

For the mixed certificate, the literal data in Appendix A specify the six lists and lower triangles of the symmetric integer matrices Z_λ . The routine `makeZ` checks the number of entries and reconstructs each

symmetric matrix. Positive definiteness is verified without eigenvalue approximation. At each step, `posdef` writes the remaining matrix as

$$B = \begin{pmatrix} C & u \\ u^\top & p \end{pmatrix}, \quad p > 40000,$$

and replaces it by the exact Schur complement $C - uu^\top/p$. The identity

$$\begin{pmatrix} x \\ t \end{pmatrix}^* B \begin{pmatrix} x \\ t \end{pmatrix} = x^*(C - uu^\top/p)x + p \left| t + \frac{u^\top x}{p} \right|^2$$

shows inductively that positive pivots prove positive definiteness, also for complex vectors. All 98 pivots in the six matrices satisfy the asserted rational inequality.

The routines `permsign` and `polynomial` implement the signed permutation sums defining Y_λ in Section 6. The convention $(P_p v)_i = v_{p(i)}$ gives $(P_p P_q v)_i = v_{q(p(i))}$, exactly `perm(perm(v,q),p)`. The routine `tagmom` uses the Gram charges $(a, -v, v - a)$ of Lemma 6.1. After multiplying the entire form by five, product assigns weight five unless both function types are one; in that case the weights for equal and different external labels are one and four. The lower-triangle summation in `gaps` doubles the off-diagonal terms. Since the polynomial coefficients and Z_λ are real, its real-moment expansion equals the full quadratic form for complex Y_λ .

Consequently the expression formed by `gaps` is exactly

$$Q = 5 \sum_{\lambda} 10^{15-e_\lambda} \mathbb{E}(Y_\lambda^* Z_\lambda Y_\lambda).$$

Apply `evmix` and then the certified mixed substitutions. On combining coefficients, we obtain the residual expansion in (41), now indexed by the program's moment keys:

$$Q = c_0 + \sum_{K \neq \emptyset} c_K \mu_K, \quad (46)$$

$$c_0 = -\frac{613302797399911}{6480}, \quad \sum_{K \neq \emptyset} |c_K| = \frac{179721388988719}{6480}.$$

Here μ_K denotes the original moment named by a retained coordinate; the notation $K \neq \emptyset$ excludes the completely empty constant key. Both equalities of rational coefficients in (46) are checked exactly by the program, as is their strictly negative sum.

It remains to justify the bound on these residual coordinates. A tagged pair coordinate is a real average of unit-modulus monomials, so $|m(A)| \leq 1$. A mixed coordinate is a real average of products of $g_s(a)$, each of absolute value at most one, so $|\mathcal{M}(\mathbf{s}; \mathbf{a})| \leq 1$. Every coordinate retained by the substitutions is one of these original moments. The new linear combinations occur as expressions in the coordinates, not as newly named coordinates to which a bound is assigned. The constant coordinate is exactly one. Hence

$$0 \leq Q \leq c_0 + \sum_{K \neq \emptyset} |c_K| = -\frac{2007321335237}{30} < 0.$$

This proves Proposition 6.3. Together with the pair coefficient check, it completes the exact verification of both certificates using only the finite equations and integer data specified in Appendix A.

A Complete exact verifier

This appendix gives the exact verifier for Propositions 3.4 and 6.3. The cubic character counts are specified by the finite sums and loops following Table 1.

The program below uses only the Python standard library and runs with Python 3.8 or later, with assertions enabled. Its standalone file is `verification/verify.py`. The accompanying wrapper `verification/run.py` checks that assertions are enabled and that the verifier has the expected checksum before running it. From the paper directory, execute

```
python3 verification/run.py
```

The complete expected output is

```
rank table: 8871 909 897
rank table: 46041 3429 2767
-613302797399911/6480 179721388988719/6480 50000000000000000
```

In each `rank table` line, the three entries give the number of nonzero equations, the number of moment coordinates, and the number of independent witness rows used for substitution, respectively. The final line gives the constant coefficient, the sum of the absolute values of all nonconstant coefficients, and the overall normalizing scale $5 \cdot 10^{15}$. Dividing the first two entries by this scale gives the constant and residual coefficient norm for $\sum_{\lambda} 10^{-e_{\lambda}} \mathbb{E}(Y_{\lambda}^* Z_{\lambda} Y_{\lambda})$. Every assertion in the program is checked in integer or rational arithmetic.

For reference, the SHA-256 checksum of `verification/verify.py` is

```
104a7306205b8544c8f913f6afdb59603f2e04a207c9ac592c383b80eb0e2aa9
```

The literal data near the end specifies each shape, its ordered list of function types and charges, its decimal scale exponent, and the lower triangle of its symmetric integer matrix, in row order.

Equation numbers in the code comments use the original numbering: the reference in `expand_degree1` is to the contractions of Lemma 3.2, and those in `mixeq` are to (13) and (14).

Listing 1: Complete exact verifier

```
1 # Exact computations on invariant moments
2 import itertools
3 from functools import lru_cache
4 from fractions import Fraction as Q
5 from math import gcd
6 from collections import Counter
7 memo = lru_cache(None)
8 memo1 = lru_cache(80000)
9
10 # tuples of tuples for matrices; "order" is the positive part norm
11 def tr(A):
12     return tuple(zip(*A))
13 def neg(a):
14     return tuple(-x for x in a)
15 def opposite(A):
16     return tuple(map(neg,A))
17 def plus(a,b):
18     return tuple(x+y for x,y in zip(a,b))
19 def order(a):
20     return sum(x for x in a if x>0)
21 def wt(L):
22     return sum(map(order,L))
23 def basic(A):
24     return tuple(sorted(a for a in A if any(a)))
25
26 @memo
27 def perms(m):
28     return tuple(itertools.permutations(range(m)))
```

```

29 def perm(a,p):
30     return tuple(a[i] for i in p)
31 @memo
32 def pp(c):
33     # Permute columns of like degree, and sort by degree
34     return tuple(p for p in perms(len(c)) if perm(c,p)==tuple(sorted(c)))
35 def canon1(A):
36     B=tr(A); c=tuple(map(order,B)); C=tuple(sorted(c))
37     return C, min(tuple(sorted(tr(perm(B,p)))) for p in pp(c))
38 @memo1
39 def canonpair(A):
40     A=basic(tr(basic(tr(A))))
41     if not A: return ()
42     B=tr(A)
43     L=[]
44     for C in (A,opposite(A),B,opposite(B)):
45         if len(C)>=len(C[0]):
46             L.append(canon1(C))
47     return min(L)[1]
48
49 @memo
50 def parts(k,n,lo,hi):
51     if not n: return ((),) if k==0 else ()
52     return tuple((i,)+L for i in range(lo,min(k//n,hi)+1) for L in parts(k-i,n-1,i,hi))
53 @memo
54 def comps(k,n,cap):
55     if not n: return ((),) if k==0 else ()
56     return tuple((i,)+L for i in range(max(0,k-sum(cap[1:])),min(k,cap[0])+1)
57                 for L in comps(k-i,n-1,cap[1:]))
58 def vecs(k,n):
59     for u in comps(k,n,(k,)*n):
60         for v in comps(k,n,tuple(k*(i==0) for i in u)):
61             yield plus(u,neg(v))
62 # sparse vectors
63 def add(d,g,s=1):
64     for a,x in g.items():
65         v=x*s+d.get(a,0)
66         if v[d[a]]=v:
67             elif a in d: del d[a]
68 def lin(g,F):
69     r={}
70     for a,x in g.items(): add(r,F(a),x)
71     return r
72
73 def expand_degree1(B,i): # sum on the right of (5), not signed
74     a=B[i]
75     C=B[:i]+B[i+1:]
76     v={}
77     for j,b in enumerate(C): add(v, evpair(canonpair(C[:j]+(plus(a,b),)+C[j+1:])))
78     return v
79
80 @memo
81 def evpair(A): # not projected yet
82     for B in (A,tr(A)):
83         assert len(B)<=6 and not any(map(sum,B))
84     for B in (A,tr(A)):
85         for i,a in enumerate(B):
86             if order(a)==1:
87                 r={}
88                 add(r, expand_degree1(B,i), -Q(1,7-len(B)))
89                 return r
90     return {A:1}
91
92 def allAB(r,c,output):
93     m=len(c); seen=set()
94     @memo1
95     def steps(i,s,t,key):
96         if i==7: return ()
97         k=r[i]; rest=sum(r[i+1:]); w=[]
98         for u in comps(k,m,s):
99             ss=plus(s,neg(u))

```

```

100         for v in comps(k,m,tuple(j if h==0 else 0 for h,j in zip(u,t))):
101             a=plus(u,neg(v))
102             if a<key:continue
103             tt=plus(t,neg(v))
104             if max(plus(ss,tt))>rest:continue
105             st=(i+1,ss,tt,a if i+1<7 and r[i+1]==k else ())
106             if i==6 or steps(*st):w.append((st,a))
107         return w
108     def visit(state,A):
109         for st,a in steps(*state):
110             C=A+(a,)
111             if len(C)<7:visit(st,C)
112             else:
113                 # uniqueness optional
114                 B=canonpair(C)
115                 if B not in seen:
116                     seen.add(B);output(C)
117         visit((0,c,c,()),())
118
119     def lcm(a,b): return a*(b//gcd(a,b))
120     def intscale(s):
121         d=1
122         for a in s.values():d=lcm(d,a.denominator)
123         return d,{i:(d//a.denominator)*a.numerator for i,a in s.items()}
124
125     def inv(a,p):return pow(a,-1,p)
126     def minusmod(s,t,b,p): # subtract b*t
127         for i,z in t.items():
128             h=(s.get(i,0)-z*b)%p
129             if h:s[i]=h
130             elif i in s:del s[i]
131
132     def echelon(E,p): # normalize at last nonzero column; wit stores source rows giving pivots
133         rel={};wit=[]
134         for vec in E:
135             s={i:x%p for i,x in vec.items() if x%p}
136             while s:
137                 k=max(s);b=s[k]
138                 if k not in rel:
139                     b=inv(b,p)
140                     rel[k]={i:z*b%p for i,z in s.items()};wit.append(vec)
141                     break
142             minusmod(s,rel[k],b,p)
143         for k in sorted(rel):
144             s=rel[k]
145             for i,z in list(s.items()):
146                 if i!=k and i in rel:minusmod(s,rel[i],z,p)
147         return rel,wit
148     def rat(z,T): # propose short rational coefficients (checked exactly later)
149         M=T; w=z
150         u,v=0,1
151         while w*w>T//2:M,w,u,v=w,M%w,v,u-M//w*v
152         return Q(w,v)
153
154     def eliminate(rows,keys=None):
155         E0=[]
156         for row in rows:
157             g=intscale(row)[1]
158             if g:E0.append(g)
159         p=10000000007
160         if keys is None:
161             keys=sorted(set(k for g in E0 for k in g))
162         num={k:i for i,k in enumerate(keys)}
163         assert len(num)==len(keys)
164         E=[{num[a]:x for a,x in g.items()} for g in E0]
165         R,small=echelon(E,p)
166         M=p
167         proj={}
168         for q in (10000000009,998244353,1000003,2013265921,754974721,32452843,86028121):
169             if proj:break
170             S,_=echelon(small,q)

```

```

171     if R.keys()!=S.keys():continue
172     iq=inv(M,q)
173     def combine(x,y):return x+((y-x)*iq%q)*M
174     for k,v in list(R.items()):
175         u=S[k]
176         R[k]={i:combine(v.get(i,0),u.get(i,0)) for i in v.keys()|u.keys()}
177     M*=q
178     rel=[]
179     for i,k in enumerate(keys):
180         if i in R:
181             f={j:-rat(z,M) for j,z in R[i].items() if j!=i}
182             if max((max(abs(x.numerator),x.denominator) for x in f.values()),default=0)**2>M//200:
183                 proj={};break
184             else:f={i:1}
185             proj[k]={keys[j]:x for j,x in f.items()}
186             rel.append(intscale(f))
187     assert proj and len(proj)==len(keys) and len(R)==len(small)
188     # Verify exactly enough equations of projection and therefore its validity
189     for g in small:
190         m=1
191         for i in g:m=lcm(m,rel[i][0])
192         s={}
193         for i,x in g.items():
194             d,f=rel[i];add(s,f,(m//d)*x)
195         assert not s
196     print("rank table:",len(E0),len(keys),len(R),flush=True)
197     return proj
198
199 def pairtable():
200     rows=[]
201     def out(A):
202         # sorted r so A[0] has order 1
203         assert order(A[0])==1 and len(A)==7
204         assert not any(map(sum,A)) and not any(map(sum,tr(A)))
205         rows.append(expand_degree1(A,0))
206     for k in range(7,12):
207         for n in range(2,min(4,k//2)+1):
208             for c in parts(k,n,2,k//2):
209                 for r in parts(k,7,1,k//2):
210                     allAB(r,c,out)
211     keys=sorted(set(k for g in rows for k in g),key=lambda A:(wt(A),A))
212     I=[0,1,2,5,4,6,7,8,9,55,80,228]
213     keys=[keys[i] for i in I]+[k for i,k in enumerate(keys) if i not in I]
214     return eliminate(rows,keys)
215
216 @memo
217 def partition(n): # equality patterns encoded by first-use indexing
218     if not n:return ((),)
219     return tuple(a+(j,) for a in partition(n-1) for j in range(max(a,default=-1)+2))
220 def fall(n,k):
221     return n*fall(n-1,k-1) if k else 1
222
223 @memo
224 def rep_pair(A): # average with replacement on the column side only
225     if not A:return lin(evpair(A),Epair.__getitem__)
226     n=len(A[0]);v={}
227     for s in partition(n):
228         k=1+max(s)
229         B=tuple(tuple(sum(x for i,x in zip(s,a) if i==j) for j in range(k)) for a in A)
230         add(v,evpair(canonpair(B)),Q(fall(6,k),6**n))
231     return lin(v,Epair.__getitem__)
232 def sortleft(A):
233     return basic(tr(basic(tr(A))))
234 def apair(*args):
235     return rep_pair(sortleft(tr(args)))
236
237 alpha=(1,1,1,-1,-1,-1)
238 def check_pair():
239     w=(-3,-2,0,1,2,2)
240     gammas=[(-1,-1,0,0,1,1),(-2,0,0,0,1,1),(-1,-1,0,1,0,1),(-1,-1,0,1,1,0)]
241     poly={(w,):-1};poly.update({(g,plus(w,neg(g))):1 for g in gammas})

```

```

242     poly[tuple([gammas[0],plus(w,neg(gammas[0]))])]=2
243     f={}
244     for a,x in poly.items():
245         for b,z in poly.items():add(f,apair(*(a+opposite(b))),3*x*z)
246     for b in ((2,0,0,-2,0,0),(2,-2,0,0,0,0)):
247         eq=f.copy();add(eq,apair(alpha,b,neg(alpha),neg(b)))
248         assert not eq
249
250 def normalize(c):
251     new={}
252     return tuple(new.setdefault(s,len(new)) for s in c)
253 @memo1
254 def cmix(c,A):
255     B=tr(A)
256     keep=tuple(i for i,b in enumerate(B) if any(b))
257     K=[]
258     for p in itertools.permutations(keep):
259         T=tr(perm(B,p))
260         K.append((normalize(perm(c,p)), min(basic(T),basic(opposite(T)))))
261     return min(K)
262 def mk(c,*args):
263     return cmix(c,basic(tr(args)))
264 def special(a):
265     return sorted(a)==sorted(alpha)
266 rows0={}
267 @memo
268 def evmix(K):
269     c,A=K
270     assert len(A)<=6 and not any(map(sum,A)) and not any(map(sum,tr(A)))
271     B=tr(A); zero=any(order(b)==1 or special(b) for b in B)
272     types=sorted(set(c))
273     if len(types)<=1:
274         v=rep_pair(A)
275         v={((),a):x for a,x in v.items()} # pairwise-average variables by special name
276         if zero:rows0[K]=v;return {}
277         return v
278     if zero:return {}
279     for i,b in enumerate(B):
280         if order(b)==2 and c.count(c[i])==1:
281             v={}
282             for t in types:
283                 if t!=c[i]:add(v,evmix(cmix(c[:i]+(t,)+c[i+1:],A)),-Q(1,7-len(types)))
284             return v
285     return {K:1}
286
287 def shift(a,i,j):
288     return tuple(x+(t==i)-(t==j) for t,x in enumerate(a))
289 def mixeq(A,rows,used,limit): # (2),(3) times a third factor, A has cols a,b,c
290     def val(S,args):
291         v=evmix(mk(S,*args));used.update(v);return v
292     for s in partition(3):val(s,tr(A)) # register zero conditions
293     for a,b,c in sorted(set(itertools.permutations(tr(A)))):
294         if a>b:continue
295         combs={(i,j):(shift(a,i,j),shift(b,j,i),c) for i in range(6) for j in range(6)}
296         if max(map(wt,combs.values()))<=limit:
297             sets= ((0,1,2),(0,1,0),(0,1,1)) if any(c) else ((0,1,0),)
298             for s in sets:
299                 v={}
300                 for (i,j),q in combs.items():
301                     if i!=j:add(v,val(s,q))
302                 if v:rows.append(v)
303     for s in (((0,0,0),(0,0,1)) if any(c) else ((0,0,0),)):
304         for i in range(6):
305             for inv0 in range(2):
306                 comb=[combs[(j,i) if inv0 else (i,j)] for j in range(6)]
307                 if max(map(wt,comb))<=limit:
308                     v={}
309                     for B in comb:add(v,val(s,B))
310                     add(v,val(s[-2:],(neg(c),c)),-1)
311                     if v:rows.append(v)
312

```

```

313 def sortedv(a):
314     return min(tuple(sorted(a)), tuple(sorted(neg(a))))
315 def triples(limit):
316     g=[list(vecs(k,6)) for k in range(limit//2+1)]
317     for n,ga in enumerate(g):
318         for a in sorted(set(map(sortedv,ga))):
319             s=set()
320             for gb in g[n+1]:
321                 for b in gb:
322                     c=neg(plus(a,b))
323                     if order(c)<=order(b) and wt((a,b,c))<=limit:s.add(tuple(sorted(zip(a,b,c))))
324             yield from sorted(s)
325
326 def mixtable(limit):
327     rows=[];keys=set()
328     for A in triples(limit):mixeq(A,rows,keys,limit)
329     for g in rows[0].values():
330         keys.update(g);rows.append(g)
331     keys=sorted(keys,key=lambda t:(bool(t[0]),wt(t[1]),t))
332     I=[0, 1, 3, 4, 5, 6, 8, 9, 10, 11, 13, 16, 20, 24, 25, 27, 31, 35, 36, 42, 44, 45, 46, 47, 57, 59,
333         60, 61, 62, 63, 64, 66, 67, 75, 76, 77, 78, 82, 85, 92, 104, 105, 123, 125, 126, 127, 128, 129,
334         132, 133, 141, 143, 144, 145, 147, 151, 153, 158, 159, 161, 162, 165, 166, 167, 172, 173, 174,
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368         3237, 3238, 3239, 3240, 3241, 3246, 3251, 3252, 3261, 3271, 3273, 3275, 3282, 3283, 3284, 3285,
369         3292, 3293, 3295, 3296, 3299, 3300, 3301, 3303, 3311, 3316, 3317, 3318, 3320, 3321, 3325, 3327,
370         3329, 3338, 3341, 3346, 3350, 3359, 3365, 3370, 3393, 3405, 3407, 3408]
371     keys=[keys[i] for i in I]+[k for i,k in enumerate(keys) if i not in I]
372     return eliminate(rows,keys)
373
374 def permsign(L):
375     pp=[itertools.permutations(s) for s in L]
376     for ls in itertools.product(*pp):
377         p=list(range(6));invn=0
378         for a,b in zip(L,ls):
379             for i,z in enumerate(b):
380                 p[a[i]]=z;invn+=sum(x>z for x in b[:i])
381             yield tuple(p), (-1)**invn
382
383 def polynomial(lam,t,v):

```

```

384     L=[];h=1
385     for r in lam:L.append(tuple(range(h,h+r)));h+=r
386     H=[tuple(s[j] for s in L if len(s)>j) for j in range(max(lam))]
387     f={}
388     for q,sgn in permsign(H):
389         for p,_ in permsign(L):
390             add(f,{(t,perm(perm(v,q),p)):sgn})
391     return f
392
393 def posdef(B):
394     B=[list(map(Q,a)) for a in B]
395     while B:
396         q=B.pop()
397         p=q.pop()
398         assert p>400000
399         B=[[x-y*a/p for x,a in zip(b,q)] for b,y in zip(B,q)]
400
401 def makeZ(m,a):
402     assert len(a)*2==m*(m+1)
403     itera=iter(a)
404     Z=[[0]*m for _ in range(m)]
405     for i in range(m):
406         for j in range(i+1):Z[i][j]=Z[j][i]=next(itera)
407     posdef(Z)
408     return Z
409
410 @memo1
411 def tagmom(X,Y): # keys for the two F types
412     s,a=X;t,v=Y;b=neg(v)
413     args=a,b,neg(plus(a,b))
414     K=mk((s,t,0),*args)
415     if not s*t:return K,None
416     return K,mk((1,2,0),*args)
417
418 def product(h,x,y,z):
419     for X,p in x.items():
420         for Y,q in y.items():
421             K,L=tagmom(X,Y)
422             s=z*p*q
423             if L is None:h[K]+=5*s
424             else:h[K]+=s;h[L]+=4*s
425
426 def gaps(data,Emix):
427     E=max(e for lam,ws,e,a in data)
428     h=Counter()
429     for lam,ws,e,a in data:
430         assert sum(lam)==5 and all(t in (0,1) and len(v)==6 and sum(v)==0 for t,v in ws)
431         B=[polynomial(lam,t,v) for t,v in ws]
432         m=len(B);Z=makeZ(m,a)
433         for i in range(m):
434             for j in range(i+1):
435                 product(h,B[i],B[j],Z[i][j]*(1+(i!=j))*10**(E-e))
436     ans=lin(lin(h,evmix),Emix.__getitem__)
437     f=ans.pop(((0,0)))
438     v=sum(abs(y) for y in ans.values())
439     print(f,v,5*10**E,flush=True)
440     assert 6480*f == -613302797399911 and 6480*v == 179721388988719
441     assert f+v<0
442
443 data=[((1, 1, 1, 2), [(0, (2, 0, -1, -2, 1, 0)), (1, (1, 0, -1, -2, 1, 1)), (1, (1, 0, -1, -2, 2, 0)),
444     (0, (1, 0, -1, -2, 1, 1)), (1, (2, 0, -1, -2, 1, 0))], 14, [7856995, -2793048, 5486181, 8971673,
445     -1693736, 37455549, 4076125, -2967341, -242661, 7655001, -23581960, 9048932, -37895129, -9254062,
446     86753990]), ((1, 1, 3), [(1, (1, 0, -1, 1, -2, 1)), (0, (3, 0, -1, -2, 0, 0)), (1, (1, 0, -1, -2,
447     0, 2)), (0, (1, 0, -3, 2, 0, 0)), (1, (3, 0, -1, -2, 0, 0)), (0, (1, 0, -1, 2, 0, -2)), (1, (1,
448     0, -1, 2, -1, -1)), (0, (1, 0, -1, -2, 0, 2)), (0, (2, 0, -3, 1, 0, 0)), (1, (1, 0, -2, 1, -1,
449     1)), (1, (1, 0, -3, 2, 0, 0)), (1, (2, 0, -3, 1, 0, 0)), (1, (2, 0, -1, -2, 0, 1)), (0, (-1, 0,
450     -1, 2, 0, 0)), (1, (1, 0, -3, 1, 0, 1)), (1, (1, 0, -1, 2, 0, -2)), (1, (2, -1, -2, 1, 0, 0)),
451     (0, (2, 0, -1, 1, 0, -2)), (1, (2, 0, -1, 1, 0, -2)), (0, (1, 0, -2, 1, 0, 0)), (1, (1, 0, -1,
452     -2, 1, 1))], 13, [8281974, 317457, 670055, -4538959, -459927, 7528052, 591578, -52372, 113157,
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